

A Compact Wideband Slot-Modified Microstrip Antenna for WiFi-6/6E/7 and Emerging 6G WLAN Systems

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Abstract— This paper proposes a new compact wide-band microstrip antenna that is an envelope-shaped radiating patch, capable of operating in frequency range of 3.3 GHz to 7.2 GHz to include WiFi-6/6E/7 and 6G WLAN frequency spectrum bands, in response to the rapidly growing need for wireless systems to provide high-throughput and low-latency data communications. The proposed antenna was developed using a 25×25 mm² area of FR-4 substrate with a partially grounded base plane, and exhibits an extremely wide impedance band (3.66–7.4 GHz) with resonant frequency of 6.69 GHz and extremely low return-loss less than -46 dB. This proposed antenna has very high radiation efficiencies greater than 90% as well as maximum gain of 4.04 dB and has low element-to-element mutual coupling less than -20 dB for a 2 × 2 MIMO antenna system which supports reliable multi-port wireless communications for modern handheld devices requiring compact size, wide band operation, and high data transfer communication capabilities.

Keywords—Microstrip antenna, ultra-wideband, WiFi-6E, WiFi-7, MIMO, WLAN 6G, CCL, Isolation.

I. INTRODUCTION

Growing demands on the increasing amount of data being transmitted at very low latencies and an ever increasing number of devices connected have driven the latest advances in modern wireless communication systems. The rapid development of wireless technology can be seen with the world-wide rollout of fifth generation (5G) mobile networks and the parallel growth of sixth generation WLANs or Wi-Fi networks; these include Wi-Fi 6 (IEEE 802.11ax) that operates in the 5.125–5.83 GHz band and supports high data rate transmission, Wi-Fi 6E (5.925–7.125 GHz), and Wi-Fi 7 (IEEE 802.11be) that will support higher data rates, higher network capacities and improved performance in a variety of dense environments [1]. The Wi-Fi industry advocated a new unlicensed spectrum allocation for instance in 2018, the FCC proposed adding 1.2 GHz in the 6 GHz band (5925–7125 MHz), effectively tripling the Wi-Fi spectrum, with final approval in 2020. Anticipating similar regulatory actions globally, the Wi-Fi ecosystem extended the IEEE 802.11ax standard to support operation in the new band. Currently, several countries, including Korea, Brazil, and Chile, permit unlicensed use across the entire 6 GHz band [2, 3]. Additionally, wireless innovation is moving closer to the Sixth Generation (6G) of cellular technology as defined by IMT-

2030; along with, the development of future WLAN standards that include Wi-Fi 7 (802.11be); and future Wi-Fi 8 (802.11bn). The development of future WLAN standards are expected to provide next generation communication systems with unprecedented levels of connectivity, reliability, and efficiency [4]. Industry forecasts indicate the development of Wi-Fi 7 (802.11be) will be a precursor to the maturity of Wi-Fi 8 (802.11bn) by 2027–28 [5]. Emerging efforts toward WLAN–6G convergence demand access to wider frequency bands for multi-gigabit data rates, but this also intensifies the challenge of designing antennas that provide wideband coverage while remaining compact and low-profile [6]

Some of the most recent designs to come out have focused on the Wi-Fi 6/6E, Wi-Fi 7, and Sub-6 GHz 5G, and all have included trade-offs in their performance in terms of growing bandwidth, gain, and size requirements for modern miniaturization age. In [7] Abdelkarim and Gharsallah described a large multiband fractal patch antenna design (75 × 57 × 1.6 mm³) that operates on 2.4, 5, and 6 GHz, and provides 6.4 dB of gain, however the trade-off here is that the design is a relatively large size. Wi et al. [8] introduced a small gap-coupled monopole antenna (50 × 30 × 1 mm³) that operates on 2.18–2.58 GHz and 5.13–7.35 GHz and has efficiencies of 70% and 67%, however gain performance and the effect of the dielectric loss were not addressed. Zhu et al. achieved a wide bandwidth (34.1%) with a slotted patch array for Wi-Fi 6/6E [9]. A single band 3.5 GHz patch antenna that exhibited a high efficiency and gain of 8.78 dBi was also demonstrated by Halder et al. [10], but lacked wideband capability for higher data transfer. Additionally, while Zhang et al. reported a wide dual-band coverage in upper band and achieved moderate gain [11]. Although superstrate-based designs by Rahman et al. [12] significantly improved impedance matching and remain narrowband. Despite these advancements, a primary limitation remains the inherent trade-off among the conflicting requirements of miniaturization, high gain, and broad bandwidth [7], [9]. Additionally, array antennas introduce the challenge of designing and integrating an efficient feeding network [10], [12].

To overcome this limitation, extensive research has examined slot-based and geometry-modification techniques [7–12]. In this context, the present work introduces a novel, compact wideband microstrip antenna that incorporates a

modified “envelope-shaped” radiating patch to achieve improved performance. This specific design was created to offer continuous wideband operation in a frequency range of 3.66 GHz to 7.4 GHz (covering the 6 GHz WLAN Band and the new Wi-Fi 6/6E/7 bands) and can be easily integrated into portable and fixed wireless applications due to its relatively simple, flat structure and high gain features. The rest of the work discusses the antenna design methodology, provides simulation-based analysis, and demonstrates its effectiveness as a high-performance solution for current 6G and next-generation WLAN applications.

II. REASEARCH METHODOLOGY

The process of designing the proposed antenna is initiated by identifying the desired frequency bands and the expected performance characteristics of WiFi-6/6E/7 and WLAN systems. Subsequent to this determination, a suitable antenna configuration and feed method are chosen, along with the selection of a substrate material that has the required dielectric properties and thickness. Initially, the antenna's size is determined via analytical formulations and then it is simulated via an electromagnetic simulation software package, such as CST. Through simulations, key performance characteristics of the antenna, including S_{11} (reflection coefficient), bandwidth, impedance matching, gain, efficiency, and radiation pattern evaluated. When the performance characteristics of the antenna do not meet the desired specifications, parameter optimization employed to improve the performance characteristics of the antenna by modifying its dimensions, adding slots or modifying the design of matching elements. After satisfactory performance characteristics have been obtained, the optimal design will be validated and compared to literature examples, and the design will be considered for future fabrication and measurement. Fig. 1 illustrates the design and optimization flow of the proposed antenna.

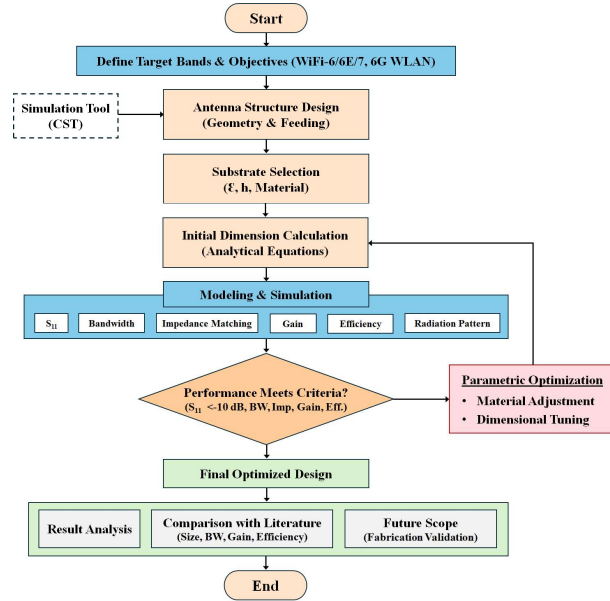


Fig. 1. Design Methodology of the Proposed Antenna

III. ANTENNA DESIGN ANALYSIS

This section describes the physical implementation and dimensional constraints of the compact, customized planar radiating structure, with the design evaluation process depicted in Fig. 2. The proposed antenna is implemented on a Rogers RT5880 substrate ($\epsilon_r = 2.2$, $\tan \delta = 0.0009$) using a square form factor measuring $25 \times 25 \times 0.79 \text{ mm}^3$. The radiating element utilizes an envelope-shaped microstrip patch with a folded inner slot to support compact and wideband operation. As shown in Fig. 3 (a-c), the single-element configuration includes the top-view layout, the ground-plane structure, and the corresponding 3D representation. The patch is excited via a 2.0 mm wide microstrip feed line to ensure that the impedance match is achieved properly at the input port from a 50Ω source. The envelope-shaped microstrip patch antenna (MPA) contains several geometric parameters which can be individually tuned to optimize the impedance bandwidth and resonant frequency characteristics. The outer arms defined as $a = 19.0 \text{ mm}$ and $b = 14.22 \text{ mm}$ are the primary radiating elements. The slot width and height within the inner section of the patch as defined by $d = 4.25 \text{ mm}$ and $e = 1.0 \text{ mm}$ have been optimized to control the current distribution and the resonant coupling between the individual sections of the patch. The lower section of the patch consists of $j = 6.90 \text{ mm}$, $k = 9.6 \text{ mm}$, and $m = 2.0 \text{ mm}$ and serves as a structural support for improving current flow continuity between the feed point and radiating area of the patch. On the backside of the substrate, a partial ground plane was also utilized, with a dimension of 19 mm in length and 5 mm in height. The implementation of the partial ground plane contributed to enhanced impedance bandwidth and radiation efficiency due to the generation of an additional resonant mode through field coupling between the patch and the truncated ground. The single element configurations of the envelope shaped radiating patches are presented in Fig. 3(a, b, c) showing the top view, the ground plane and three dimensional views of the antenna. The 2×2 MIMO antenna configuration of the single element design is provided in Fig. 3(d). The high throughput requirements of Wi-Fi 5/6 and 6G applications were addressed by this MIMO antenna configuration of size $50 \times 50.6 \text{ mm}^2$. The impedance characteristics of the single element design in terms of the reflection coefficient are presented in Fig. 4 to demonstrate the impedance matching quality and resonant behavior during the development of the antenna design. The antenna dimensional parameter listed in Table I, have been carefully optimized using full-wave electromagnetic simulation to achieve desired performance characteristics, such as wide impedance bandwidth, good return loss, and stable radiation patterns.

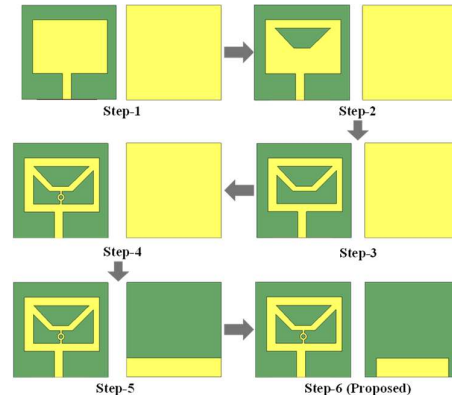


Fig. 2. Design and evaluation process of the proposed MPA.

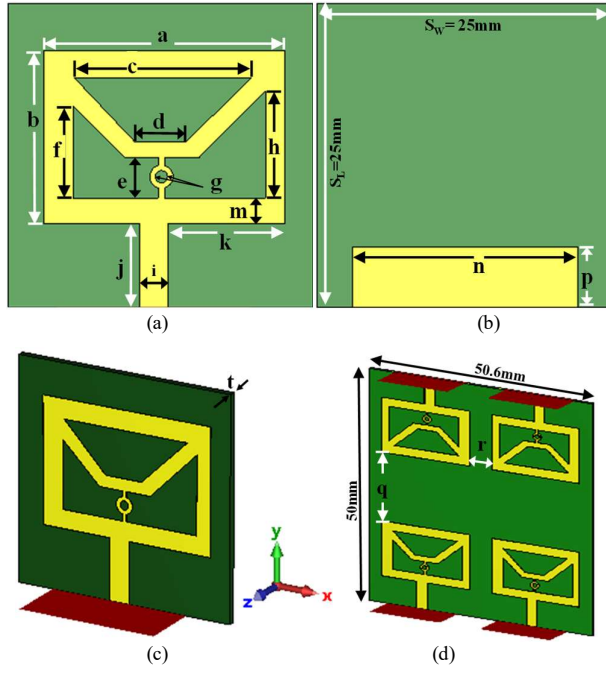


Fig. 3. Geometry of the proposed compact MPA (a) single element front view, (b) single element back view, (c) single element 3D view and (d) 2x2 MIMO antenna 3D view.

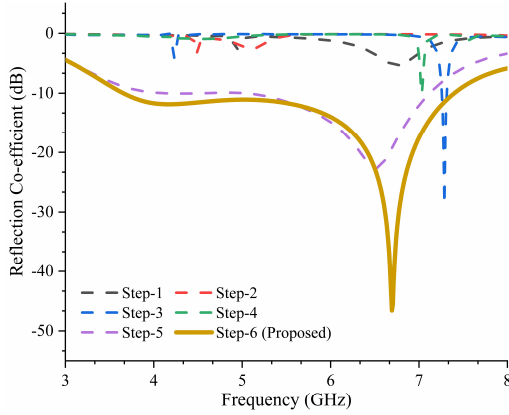


Fig. 4. S_{11} comparison across design iterations.

TABLE I. KEY GEOMETRICAL DIMENSIONS OF THE PROPOSED

Parameter	Unit (mm)	Parameter	Unit (mm)	Parameter	Unit (mm)
a	19.00	g	1.00	n	19
b	14.22	h	8.90	p	5
c	19.80	i	2.30	t	0.79
d	4.25	j	6.90	q	17.76
e	3.30	k	9.6	r	5.20
f	7.60	m	2.00	$S_L = S_W$	25

IV. ANTENNA PERFORMANCE ANALYSIS

The performance of the proposed antenna was thoroughly assessed by utilizing a Finite Element Method (FEM) solver within CST Microwave Studio. Finally, CST Microwave Studio simulated data included all final simulated parameters such as impedance bandwidth, return loss, radiation patterns, peak gain, and radiation efficiencies were evaluated to determine whether it is suitable for use across the Wi-Fi-6/6E/7 and WLAN 6G spectrums.

A. Reflection Co-efficient

The reflection coefficient (S_{11}) plot shown in Fig. 5 clearly demonstrates that the antenna has achieved a continuous Ultra-Wideband (UWB) impedance match over the entire operating frequency band of interest with a magnitude less than -10 dB, which is well beyond the standard ultra-wideband requirements of approximately 3.66 GHz through 7.4 GHz. This wide matching bandwidth successfully encompasses the high-capacity C-Band and the entirety of the 6 GHz WLAN spectrum, making the design highly suitable for supporting Wi-Fi 7 as well as the WLAN/6E/6G band (5.925–7.125 GHz). Furthermore, the extremely high resonance of approximately -46.24 dB at 6.69 GHz, suggests the antenna is heavily optimized for peak efficiency within the 6 GHz band. While this strong S_{11} performance validates the potential for high radiation efficiency.

With a 2x2 MIMO port configuration the reflection coefficient characteristics of all four ports are consistent with being <-10 dB throughout the operational band and achieve a minimum of -65 dB at resonance, therefore demonstrating excellent impedance matching as illustrated in Fig. 6. The plot of the mutual coupling between each element of the MIMO array illustrated in Fig. 7 demonstrates the loss of signal between elements is greater than -20 dB at all frequencies within the designated ultra-wideband operating range and will therefore provide sufficient isolation between elements to minimize the effect of cross-talk, preserve the diversity aspects of the MIMO system and allow for the high capacity data rates required by many advanced wireless communications systems utilizing this type of antenna technology.

B. Voltage Standing Wave Ratio (VSWR) and Impedance matching, TARC, ECC

The data presented in Fig. 8 and Fig. 9 offer a complementary understanding of the impedance matching and operational bandwidth of antenna's, which are foundational requirements for its suitable applications. The VSWR plot (Fig. 8) confirms that the antenna successfully achieves an Ultra-Wideband (UWB) match, maintaining performance below the threshold ($VSWR \leq 2.0$). The VSWR reaches an optimal match, dropping near 1.009 at 6.69 GHz, indicating a peak resonance point precisely within the high-priority WLAN band. Supporting this observation, the impedance plot (Fig. 9) shows that in the region around 6.69 GHz

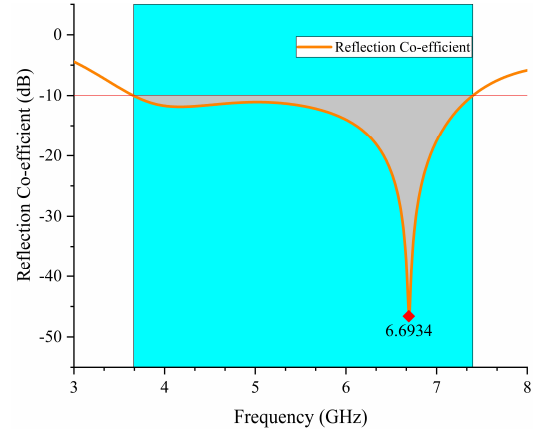


Fig. 5. S_{11} response of the proposed single element MPA.

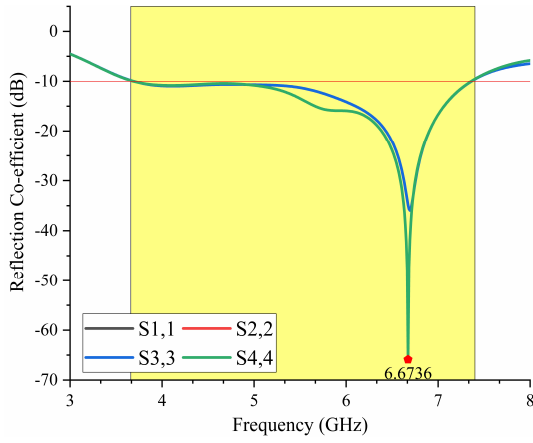


Fig. 6. S-parameter characteristics (S11, S22, S33, S44) of the proposed MIMO antenna.

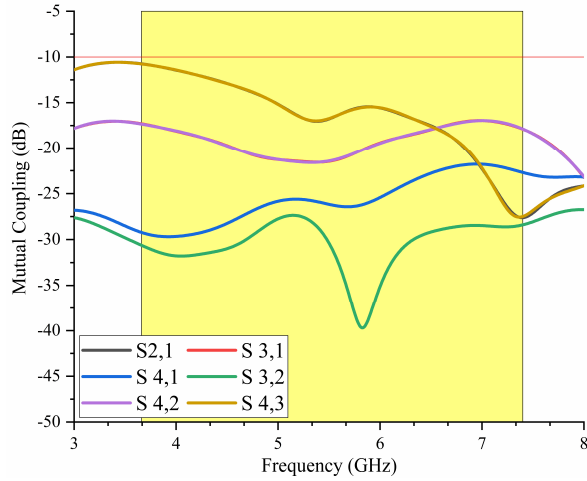


Fig. 7. Mutual coupling (S21, S31, S41, S32, S42, S43) characteristics of the proposed MIMO antenna.

the imaginary part (reactance) closely approaches $j0.364 \Omega$, which defines resonance, while the Real Part (resistance) is near 50.02Ω , confirming the excellent impedance transformation at that point.

The strong UWB performance (low VSWR/high efficiency) across the bands enables high-performance MIMO operation by maintaining low Total Active Reflection Coefficient (TARC) values -12 dB near 7 GHz and ensuring

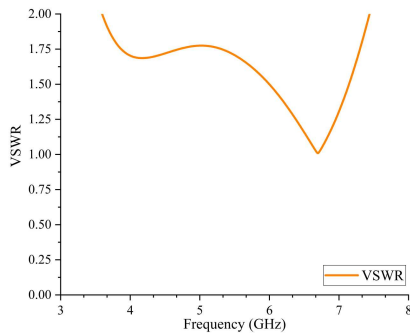


Fig. 8. VSWR response of the proposed single element MPA.

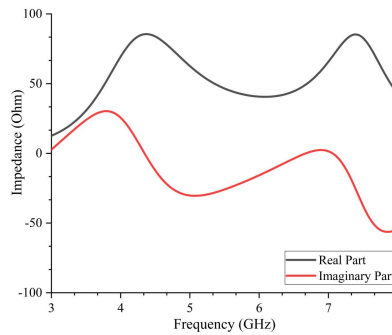


Fig. 9. Input impedance characteristics of the proposed single MPA.

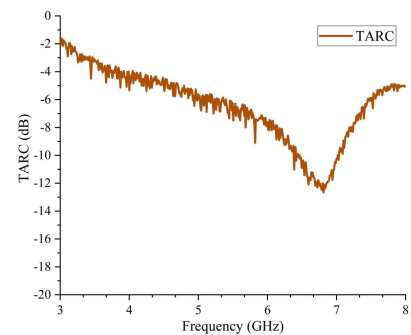


Fig. 10. TARC response of the proposed MIMO antenna.

excellent diversity behavior (Fig. 10), with Envelope Correlation Coefficients (ECC) remaining below 0.02 and in some cases nearly zero when effective decoupling structures are employed (Fig. 11).

C. Radiation Performance, CCL

This UWB match is accompanied by exceptionally high radiometric performance as shown in Fig. 12 the Radiation Efficiency consistently exceeds 85% across the entire band, peaking at approximately 91% in the 6 GHz range, which validates the use of low-loss materials critical for minimizing power consumption and heat dissipation. The peak of the directivity for Fig. 13 is 4.39 dBi, this peak corresponds to a realized gain of 4.04 dB in the 6G WLAN band and a 3D far-field radiation pattern (at 6.6934 GHz) in Fig. 14 (a, b) with a quasi-omni directional radiation pattern that will provide adequate signal coverage over a large angular range.. As shown in Fig. 15, the Channel Capacity Loss (CCL) of the proposed 2×2 MIMO antenna remains well below the accepted 0.4 bits/s/Hz limit across the operating band, demonstrating high radiation efficiency and low correlation among the antenna elements key indicators of strong MIMO channel capacity.

D. Electrical, Magnetic Field and Surface Current Distribution

Radiation behavior of the proposed antenna is depicted by Fig. 16 in terms of both E- and H-plane field distributions for $\phi = 0^\circ$ and $\phi = 90^\circ$ at the resonance frequency of 6.6934 GHz. Magnitude of the E-field pattern in the main lobe at $\phi = 90^\circ$ was 18.7 dB(V/m) and at $\phi = 0^\circ$ was 18 dB(V/m). Directions of main lobes were 19° and 7° , while corresponding 3 dB beamwidths were 73.2° and 88.1° , respectively. H-plane field plot has shown magnitude of the main lobe as -32.8 dB(A/m) at $\phi = 90^\circ$ and -33.5 dB(A/m) at $\phi = 0^\circ$; they have similar orientation of beams and width of angles. Therefore, these almost symmetrical and stable radiation profiles in both planes illustrate balanced electromagnetic field distribution of the antenna as well as consistent radiation properties of the antenna at its operation frequency. Thus, this illustrates that the antenna will support efficient far-field radiations with a good directivity and low level of cross-polarizations for WLAN/6G (5.925–7.125 GHz) applications as well as WLAN bands. Directional beam characteristics and wide angle of coverage indicate reliable and effective propagation in high frequency wireless systems.

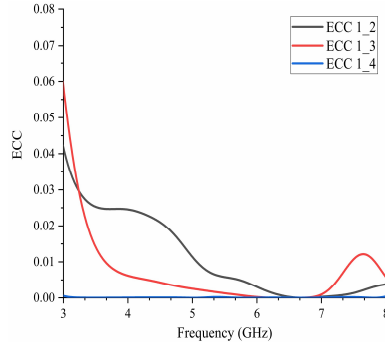


Fig. 11. Correlation performance (ECC) of the proposed MIMO antenna.

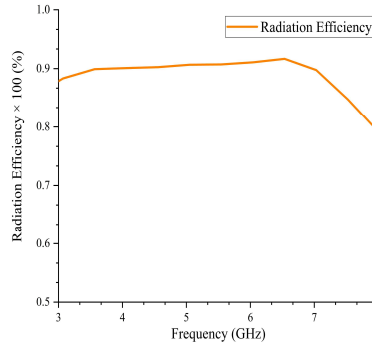


Fig. 12. Radiation efficiency performance across the operating band for the proposed MPA.

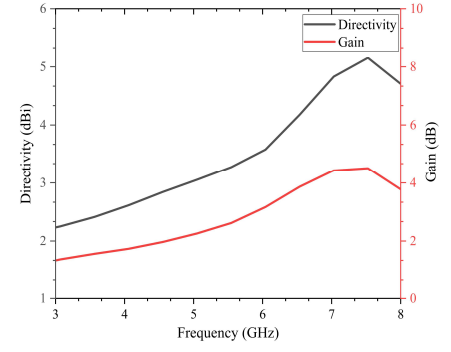


Fig. 13. Frequency response analysis of directivity and gain (IEEE) for the proposed MPA.

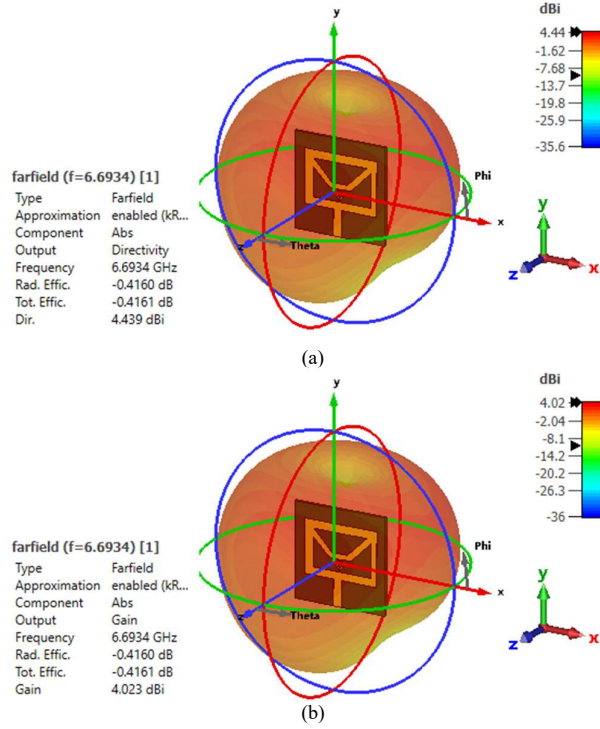


Fig. 14. 3D radiation characteristics (a) directivity and (b) gain of the proposed MPA.

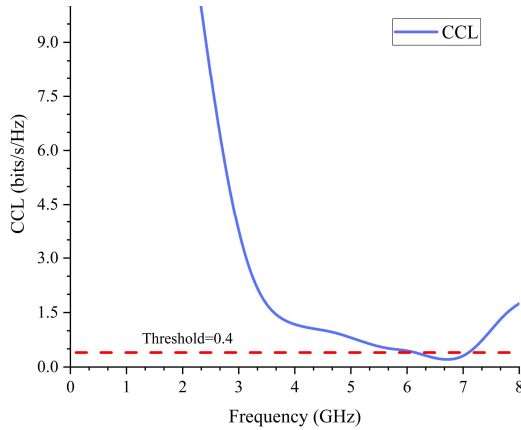


Fig. 15. Response of channel capacity loss (CCL) for the proposed 2x2 MIMO antenna.

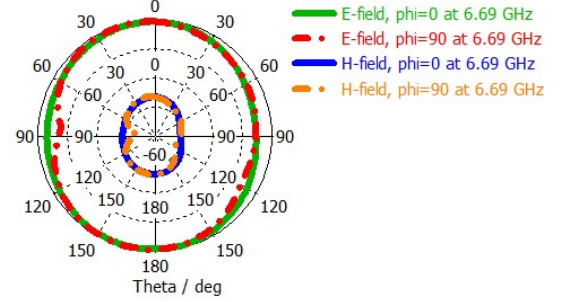


Fig. 16. E-field and H-field radiation patterns at $\phi = 0^\circ, 90^\circ$ for the proposed single element MPA at resonance frequency 6.69 GHz.



Fig. 17. Surface current distribution of the proposed antenna at 6.69 GHz.

The surface current distribution for the proposed microstrip patch antenna is illustrated in Fig. 17 at the resonant frequency of 6.6934 GHz where the maximum surface current reaches approximately 101.912 A/m. Strong currents exist at the feedline and the bottom portion of the patch, showing that both of these patches are being utilized for the energizing of the antenna. The currents flow upwards, and are spread out over the entire area of the major radiating patch, creating two identical paths through which the current flows from the slots at the edges of the patch, and around the top of the corner. A complete summary of the results obtained during testing are shown in Table II.

TABLE II. KEY FINDINGS AND PERFORMANCE PARAMETERS OF THE PROPOSED ANTENNA AT RESONANCE FREQUENCY

Description	Value
Overall dimension (mm ³)	25 × 25 × 0.79
Frequency range (GHz)	3.66-7.4
Resonance frequency (GHz)	6.69
VSWR	1.009
Maximum radiation efficiency (%)	90.91
Maximum directivity (dBi)	4.39
Maximum gain (dB)	4.04
Impedance (Ω)	50.02 ± 0.364

V. COMPARATIVE ANALYSIS

The comparative assessment indicates that the proposed antenna offers meaningful enhancements relative to existing designs. While several referenced works reported in [7]–[12], which typically exhibit larger physical sizes, limited bandwidths, or moderate gain and efficiency, the proposed design demonstrates several notable improvements. Existing

works provide bandwidths ranging from 100-400 MHz, moderate gain, and efficiencies that are often unreported or below 90%. In contrast, the proposed antenna achieves a compact footprint of $25 \times 25 \times 0.79$ mm³ while delivering a significantly wider 3.74 GHz bandwidth and a stable resonance at 6.69 GHz, suitable for Wi-Fi-6/6E/7 and WLAN 6G applications.

TABLE III. PERFORMANCE COMPARISON BETWEEN THE PROPOSED ANTENNA AND RECENT STUDIES.

Ref.	Size (L×W×h) mm ³	Substrate	Operating Frequency Range (GHz)	BW (GHz)	Resonance Frequency (GHz)	Reflection Co- efficient (dB)	Gain (dB)	Efficiency (%)
[7]	57×75×1.6	Rogers RT 5880	2.46–2.48, 5.1–5.2, 5.9–6.0	200, 100, 100	2.4, 5, 6	-30, -33, -27.5	6.4, 6.3, 3	–
[8]	50×30×1	FR-4	2.18–2.55, 5.15–7.125	400, 2220	2.5, 5.4	-16, -20, -15	6	70, 67
[9]	29 × 32 × 3	F4B	5.1–7.2	2100	5.5, 6, 6.65	~35, -21.75, -32	14.1	53-77.6
[10]	100×70×0.6	Rogers RT 5880	~3.45–3.55	100	3.5	-34.98	8.78	89.87
[11]	38×26×0.8	FR-4	2.43–2.72, 3.31–7.32	290, 4010	2.6, 3.5, 4.9, 6.7	-15, -20, -23, -16	5.48	~95
[12]	24.35×20.51×1.5	FR-4	5.58–5.84	260	5.71	-75.69	2.07	–
Proposed work	25×25×0.79	Rogers RT 5880	3.66–7.4	3740	6.69	-46.24	4.04	90.91

VI. CONCLUSION

The proposed envelope-shaped slot-based microstrip antenna achieves a significant advancement in compact wideband antenna design for next-generation WLAN and wireless communication systems. The antenna's ultra-wide bandwidth from 3.66 GHz to 7.4 GHz fully covers the emerging Wi-Fi-6/6E/7 and WLAN 6G bands with stable resonance and strong impedance matching. High radiation efficiency over 90%, a peak gain exceeding 4 dB, and excellent isolation (<-20 dB) among antenna elements in the MIMO configuration confirm its potential for high-performance wireless applications. The channel capacity loss (CCL) remains well below 0.4 bits/s/Hz, total active reflection coefficient (TARC) values are maintained below -12 dB near 7 GHz, and the envelope correlation coefficient (ECC) values stay under 0.02, indicating strong MIMO channel capacity and diversity performance. Compared with contemporary designs, this solution offers a superior balance of compact footprint, bandwidth, gain, and efficiency, making it well-suited for integration into portable and fixed wireless devices targeting multi-gigabit throughput and low-latency communications.

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